DARPA Triage Drone

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Abstract

The DARPA Triage Drone is an unmanned aerial system designed to assess and prioritize disaster victims without placing first responders in harm's way. Inherited from a Fall 2024 capstone team, the project aimed to integrate sensing, computing, communication, and power hardware onto the Aurelia X4 Standard quadrotor to create an airborne triage unit capable of measuring vital signs, evaluating cognitive responsiveness, and relaying findings to an operator in real time. Due to procurement constraints, the drone was never received, and the team's primary deliverable shifted to laying the groundwork for a future implementation. The centerpiece of this effort is a fully designed and 3D-printed belly pack — a modular payload housing engineered to stay within the Aurelia X4's 1.5 kg payload limit, maintain center-of-gravity balance, and provide mounting locations for all triage hardware.

Subsystem Overviews

Power & Safety Subsystem — Power from the Aurelia X4's 22.2V 10,000 mAh LiPo battery is split via a terminal block to feed the drone and a voltage converter stepping down to 12V for the Jetson Nano. A 60A fuse and reverse polarity protection guard all circuits. The added electronics account for only 4.7% of total power draw, preserving an estimated 20 minutes of flight time.

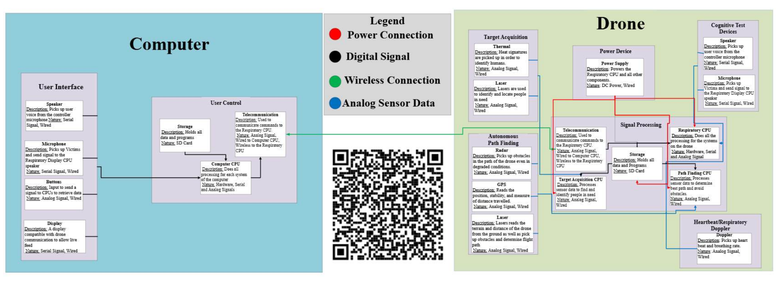
Weight Distribution & Sensor Placement Subsystem — All triage hardware mounts into a custom 3D-printed belly pack attached to the Aurelia X4 using existing mounting points, requiring no structural modifications. Heavy components are centered to keep CG shift within 5% of the motor-to-motor span, with total payload targeting ≤1.35 kg — a 10% margin below the 1.5 kg limit.

Signal Processing Subsystem — The Infineon BGT60UTR11AIP 60 GHz radar's microcontroller processes raw data through a multi-stage FIR filter pipeline, isolating respiration (0.13–1.0 Hz) and heart rate (0.8–3.0 Hz). FFT peak detection with a 6 dB SNR threshold converts dominant frequencies to BPM within 500 ms. Python simulations validated the pipeline, accurately recovering various test heart and breath rates.

Communication Subsystem — A ReSpeaker microphone array, dual cameras, and two 3W speakers are managed by the Jetson Nano via Python. Propeller noise is suppressed by a software adaptive notch filter using frame-by-frame FFT analysis, while a fixed bandpass filter preserves the 100–4,000 Hz speech range. Microphone and speaker can operate in half-duplex to prevent feedback during cognitive assessment.

Drone-to-PC Link Subsystem — The Jetson Nano hosts a dedicated Wi-Fi hotspot that the operator's laptop connects to directly. A 5 GHz 802.11ac adapter streams video, audio, and vitals from the Jetson Nano to the laptop via WebRTC with DTLS/SRTP encryption. All data is discarded immediately after display per DHA 6000.02.

Hardware Block Diagram



Project Photo

Acknowledgements

- TTU ECE Department
- Dr. Christopher Storm Johnson
- Previous DARPA Triage Drone Team
- Dr. Dale Blair

Testing

Vitals Detection — The radar sensor was tested for heart rate and breathing rate detection across two subjects at 0.5 m and 1.0 m. Heart rate accuracy was poor, with mean errors exceeding 48% at both distances due to the algorithm locking onto harmonic frequencies rather than the fundamental rate. Breathing rate performed better at 21.25% mean error at 1.0 m, though accuracy unexpectedly degraded at closer range. Both metrics were detected within their specified frequency bands, confirming the hardware can sense physiological motion at the required distance, but algorithm refinement is needed before operational deployment.

Adaptive Notch Filter — The adaptive notch filter was tested using a ReSpeaker microphone array on the Jetson Nano, with simulated drone noise played through a smartphone speaker. The filter uses frame-by-frame FFT analysis to detect dominant rotor noise frequencies and dynamically center notch filters to suppress them. During testing, the filter locked onto a stable frequency of 148 Hz within one reading and maintained it for 13 consecutive frames. When the simulated rotor speed changed, the filter adapted and stabilized at new frequencies of 356 Hz and 380 Hz respectively. Human speech frequencies (300–4,000 Hz) were preserved throughout, consistent with design specifications. Full validation with actual drone hardware is recommended before field deployment.

BOM

BILL OF MATERIALS			
Description	Qty	Unit Cost	Total Cost
4.5V–28V 6A Synchronous Buck Converter	1	$2.63	$2.63
22.2V 6S LiPo Battery 25C 10000mAh	1	$158.14	$158.14
Intel Dual Band Wireless-AC 8265 M.2 Wifi Card	1	$17.09	$17.09
IPEX UFL to RP-SMA Female Bulkhead Pigtail	1	$7.99	$7.99
Omnidirectional RP-SMA Dual Band Antenna	1	$6.99	$6.99
Reverse Polarity Protection Circuit, 60A	1	$11.50	$11.50
Dual Row Screw Terminal Block 600V 60A 4-Pos	1	$24.38	$24.38
Heart and Respiratory Rate Monitor	1	$39.99	$39.99
Microphone Windproof Windscreen	1	$5.98	$5.98
Microphone Array & Audio Filtering Chip	1	$100.76	$100.76
Wide Angle 15-pin Camera (IMX219)	2	$24.99	$49.98
500W 1 Ohm Resistor	1	$74.91	$74.91
Zip Ties 8-inch, 100 Pack	1	$4.99	$4.99
Velcro ONE-WRAP Roll, 12ft × 3/4in	1	$6.97	$6.97
PETG 3D Printer Filament 1.75mm	1	$28.49	$28.49
LiPo/Li-Ion Balance Charger	1	$43.99	$43.99
		TOTAL	$584.78

Future Work

With the foundational hardware design, signal processing pipeline, and communication architecture fully documented, the primary next step is procurement and physical integration of the Aurelia X4 drone with the completed belly pack. Further development should focus on refining the heart rate peak detection algorithm to suppress breathing harmonic interference and improving breathing rate consistency across subjects and distances. Full system validation, including in-flight vitals testing and real rotor noise filtering, would complete the transition from proof of concept to a deployable triage platform.